Basic Differentiation

Section
2.3

A. Properties and Formulas (The short way—Yeah!)

1. Basic Functions

Function	Derivative
$f(x)=c$ (Constant)	$f'(x)=0$
$f(x)=x$	$f'(x)=1$
$f(x)=cx$	$f'(x)=c$
$f(x)=x^n$	$f'(x)=nx^{n-1}$
$f(x)=cx^n$	$f'(x)=cnx^{n-1}$
$f(x)=c\cdot g(x)$	$f'(x)=c\cdot g'(x)$
$f(x)=g(x)+h(x)$	$f'(x)=g'(x)+h'(x)$
$f(x)=g(x)-h(x)$	$f'(x)=g'(x)-h'(x)$

Note: For the function $f(x)=g(x)\cdot h(x)$ we CANNOT say that $f'(x)=g'(x)\cdot h'(x)$

For the function $f(x)=\dfrac{g(x)}{h(x)}$ we CANNOT say that $f'(x)=\dfrac{g'(x)}{h'(x)}$

2. Trigonometric Functions

Function	Derivative
$f(x)=\sin x$	$f'(x)=\cos x$
$f(x)=\cos x$	$f'(x)=-\sin x$
$f(x)=\tan x$	$f'(x)=\sec^2 x$
$f(x)=\csc x$	$f'(x)=-\csc x\cot x$
$f(x)=\sec x$	$f'(x)=\sec x\tan x$
$f(x)=\cot x$	$f'(x)=-\csc^2 x$

Examples: Find and LABEL the derivatives of each of the following functions.

1. **(video)** $f(x)=17$

2. **(video)** $y=3x$

3. **(video)** $k(x)=x^5$

4. **(video)** $h(x)=5x^3$

5. **(video)** $s(r)=4\pi r^2$

6. **(video)** $\dfrac{d}{dx}\left(\sqrt{x}\right)$

7. **(video)** $\dfrac{d}{dx}\left(4x^{\frac{3}{2}}\right)$

8. **(video)** $v(t)=\dfrac{1}{t}$

Common Derivatives

$$\frac{d\left(\sqrt{x}\right)}{dx}=\frac{1}{2\sqrt{x}} \qquad \text{Since} \qquad \frac{d\left(\sqrt{x}\right)}{dx}=\frac{d\,(x^{\frac{1}{2}})}{dx}=\frac{1}{2}x^{-\frac{1}{2}}=\frac{1}{2\sqrt{x}}$$

$$\frac{d\left(\frac{1}{x}\right)}{dx}=-\frac{1}{x^2} \qquad \text{Since} \qquad \frac{d\left(\frac{1}{x}\right)}{dx}=\frac{d\left(x^{-1}\right)}{dx}=-x^{-2}=-\frac{1}{x^2}$$

9. **(video)** Find the derivative $d(x)=\frac{9}{x^3}+3x+2$

10. **(video)** Find the derivative $g(t)=5t^2+\frac{4}{t^2}-18$

11. **(video)** Find the derivative $g(t)=4t^3-\sqrt{t}+2$

12. $p(x)=\frac{x^3+4x^2}{x}$

13. $d(x)=(x+3)^2$

14. $f(x)=\sin x+\tan x$

15. $g(x)=2\cos x+\sec x$

16. $f(x)=\sqrt[3]{x}-\cot x$

More Examples

17. $f(x)=x^3+4x^2-2x+4$

Find $f'(x)=$

$f'(2)=$

Find $f''(x)=$

$f''(2)=$

18. $g(t)=\dfrac{4t^2+t+5}{\sqrt{t}}$

Find $g'(t)=$

$g'(1)=$

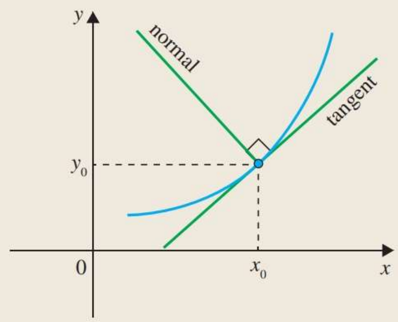
Find $g''(t)=$

$g''(1)=$

19. $h(x)=x^5-2x^4+3x^3-x-6$ Find the first six derivatives of the function.

B. Normal and Tangent Lines to a Function

A line that is perpendicular to the tangent line at the point of intersection



Examples

20. Find the horizontal tangent lines (lines with slope = 0) to the function $f(x) = 2x^3 + 3x^2 - 120x + 23$

21. Find the tangent and the normal lines to the function $f(x) = 4\cos x$ at $x = \dfrac{\pi}{3}$

C. Applications to Position, Velocity, and Acceleration

If the motion/position function of a particle is known, we can find the velocity and acceleration functions in the following way.

- If the **position** of a particle is given by $f(x)$, then the **velocity** of the particle is given by $f'(x)$
- If the **velocity** of a particle is given by $g(x)$, then the **acceleration** of the particle is given by $g'(x)$

  (We can also say that if the position of a particle is given by $f(x)$, then the acceleration of the particle is given by $f''(x)$, the second derivative of the motion function.)

Alternative notation:

- **Position** of a particle $s(t)$
- **Velocity** of a particle $v(t)$
- **Acceleration** of a particle $a(t)$

Then $v(t) = s'(t)$ and $a(t) = v'(t) = s''(t)$

Example

22. A particle's **position** is described by the function $s(t) = 3t^3 - 144t$ (t is measures in seconds and $s(t)$ in feet.)

a. Find the velocity function.

b. Find the acceleration function.

c. Find the acceleration after 9 seconds.

d. Find the acceleration when the velocity is 0.

23. A particle's **position** is described by the function $s(t)=t^3-9t^2+15t+10$ (t is measures in seconds and $s(t)$ in feet.)

a. Find the velocity function.

b. What is the velocity after 3 seconds?

c. When is the particle at rest?

d. When is the particle moving in a positive direction?

e. When is the particle slowing down?

f. Find the total distance traveled during the first 8 seconds.

More Examples (Webwork)

24. The area of a disc with radius r is $A(r)=\pi r^2$. Find the rate of change of the area of the disc with respect to its radius when $r=5$.

25. If $f(x)=\left(\frac{3}{4}x\right)^9$, then $f'(x)=$

26. If $f(x)=\sqrt{x}\cdot(3x+2)$, then $f'(x)=$

27. **a.** If $f(x)=12\pi^2$, then $f'(x)=$

b. $f(x)=12x^2$, then $f'(x)=$

c. $f(x)=12\pi x^2$, then $f'(x)=$

28. If a ball is thrown vertically upward from the roof of 64-foot building with a velocity of 32 ft/sec, its height after t seconds is $s(t) = 64 + 32t - 16t^2$

a. What is the maximum height the ball reaches?

b. What is the velocity of the ball when it hits the ground (height 0)?